# I- Reciprocal Trigonometric Functions

**Definition**

We define the functions *cosec* x, *sec* x, and *cot* x as the **reciprocal** functions of $\sin x$, $\cos x$, and $\tan x$ respectively :

$$\text{cosec } x = \frac{1}{\sin x}, \qquad \sec x = \frac{1}{\cos x}, \qquad \text{and} \qquad \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x}.$$

**Example**

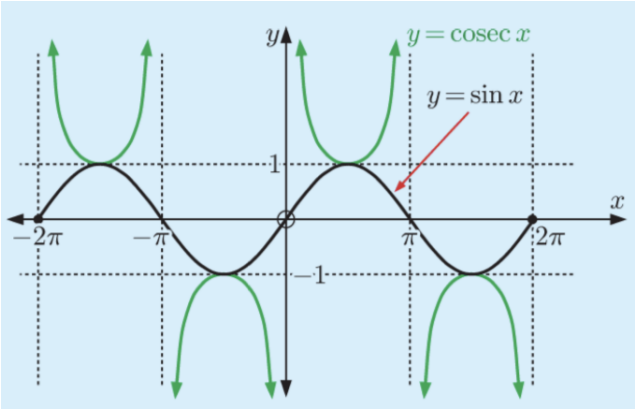
$$\text{cosec } \frac{\pi}{6} = \frac{1}{\sin \frac{\pi}{6}} = \frac{1}{\frac{1}{2}} = 2.$$

**Example**

Using the graph of $y = \sin x$, sketch the graph of

$$y = \frac{1}{\sin x} = \text{cosec } x \quad \text{for } -2\pi \leqslant x \leqslant 2\pi.$$

Check your answer using technology.



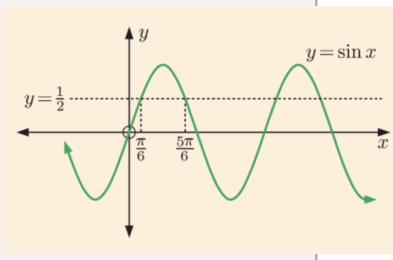

# II- Inverse Trigonometric Functions

In many problems we need to know what angle results in a particular trigonometric ratio. We have previously seen this for right-angled triangle problems and when using the cosine and sine rules.

Rather than writing the inverse trigonometric functions as $\sin^{-1}$, $\cos^{-1}$, and $\tan^{-1}$, which can be confused with reciprocals, mathematicians more formally refer to these functions as the **inverse trigonometric functions** *arcsine*, *arccosine*, and *arctangent*.

## Investigation

The function $y = \sin x$ fails the horizontal line test on its natural domain. For example, when $y = \frac{1}{2}$, there is more than one corresponding value of $x$, since $\sin x = \frac{1}{2}$ has infinitely many solutions, including




$$x = \frac{\pi}{6} \quad \text{and} \quad x = \frac{5\pi}{6}.$$

Therefore, if we want to find inverse functions for $\sin x$, $\cos x$, and $\tan x$, we will need to apply suitable domain restrictions.

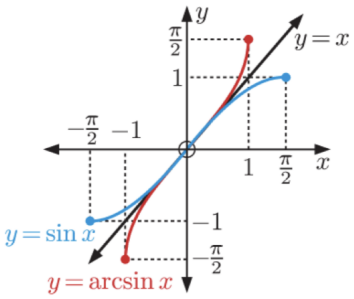
1. Draw the graphs of $y = \sin x$, $y = \cos x$, and $y = \tan x$ on the domain $-3\pi \leqslant x \leqslant 3\pi$.
2. Complete the following table, indicating whether each function is one-to-one on the given restricted domain.

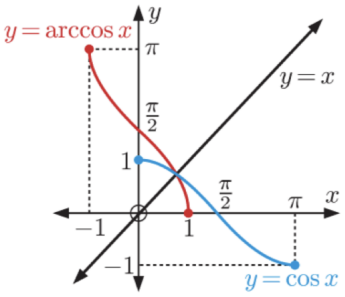
| **Restricted domain** | $\sin x$ | $\cos x$ | $\tan x$ |
|---|---|---|---|
| $0 \leqslant x \leqslant 2\pi$ | | | |
| $-\pi \leqslant x \leqslant \pi$ | | | |
| $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$ | | | |
| $0 \leqslant x \leqslant \pi$ | | | |
| $-\frac{\pi}{2} \leqslant x \leqslant \frac{3\pi}{2}$ | | | |
| $\pi \leqslant x \leqslant 2\pi$ | | | |

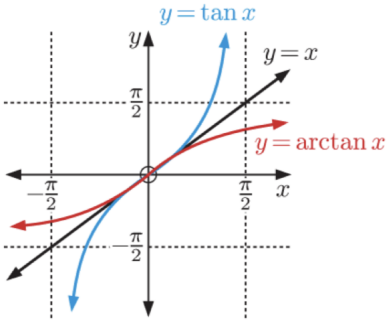
3. Discuss which domain restriction you think would be most suitable for the inverse function of :
   (a) $\sin x$
   (b) $\cos x$
   (c) $\tan x$
4. The function $f(x) = \sin x,\ -\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$ has inverse $f^{-1}(x) = \arcsin x$. Sketch the graph of $y = f(x)$, and hence sketch the graph of $f^{-1}(x) = \arcsin x$ on the same set of axes.
5. The function $f(x) = \cos x,\ 0 \leqslant x \leqslant \pi$ has inverse $f^{-1}(x) = \arccos x$. Sketch the graph of $y = f(x)$, and hence sketch the graph of $f^{-1}(x) = \arccos x$ on the same set of axes.
6. The function $f(x) = \tan x,\ -\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$ has inverse $f^{-1}(x) = \arctan x$. Sketch the graph of $y = f(x)$, and hence sketch the graph of $f^{-1}(x) = \arctan x$ on the same set of axes.

## Definition

| Inverse trigonometric function | Definition | Range |
|---|---|---|
| $y = \arcsin x$ | $x = \sin y, \quad -1 \leqslant x \leqslant 1$ | $-\frac{\pi}{2} \leqslant y \leqslant \frac{\pi}{2}$ |
| $y = \arccos x$ | $x = \cos y, \quad -1 \leqslant x \leqslant 1$ | $0 \leqslant y \leqslant \pi$ |
| $y = \arctan x$ | $x = \tan y, \quad x \in \mathbb{R}$ | $-\frac{\pi}{2} < y < \frac{\pi}{2}$ |




## Example

Find, where possible, the exact solutions of :

1. $\arctan x = \frac{\pi}{3}$
2. $\arccos(x-1) = \frac{2\pi}{3}$
3. $\arcsin x = \frac{2\pi}{3}$

# III- Algebra and Trigonometric Functions

**Example**

Simplify :

1. $3\cos\theta + 4\cos\theta$
2. $\tan\alpha - 3\tan\alpha$

**Property**

Some angles relationships :

The negative angle formulae are established by reflection in the $x$-axis :

$$\sin(-\theta) = -\sin\theta \qquad \cos(-\theta) = \cos\theta$$

The supplementary angle formulae are established by reflection in the $y$-axis :

$$\sin(\pi - \theta) = \sin\theta \qquad \cos(\pi - \theta) = -\cos\theta$$

The complementary angle formulae are established by reflection in the line $y = x$ :

$$\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta \qquad \cos\left(\frac{\pi}{2} - \theta\right) = \sin\theta$$

**Example**

Simplify :

1. $\sin(-\theta) + 2\sin\theta$
2. $\dfrac{\cos\left(\frac{\pi}{2} - \theta\right)}{\cos(\pi - \theta)}$

**Property**

The Pythagorean identity is : $\sin^2\theta + \cos^2\theta = 1$

**Property**

$\tan^2\theta + 1 = \sec^2\theta$ and $1 + \cot^2\theta = \csc^2\theta$

**Proof**

Using $\sin^2\theta + \cos^2\theta = 1$, $\frac{\sin^2\theta}{\cos^2\theta} + \frac{\cos^2\theta}{\cos^2\theta} = \frac{1}{\cos^2\theta}$ $\Rightarrow$ $\tan^2\theta + 1 = \sec^2\theta$

**Example**

Simplify :

1. $2 - 2\sin^2\theta$
2. $\cos^2\theta\sin\theta + \sin^3\theta$

**Example**

Expand and simplify :

$(\cos\theta - \sin\theta)^2$

**Example**

Factorise :

1. $\cos^2\alpha - \sin^2\alpha$
2. $\tan^2\theta - 3\tan\theta + 2$

**Example**

Simplify :

1. $\dfrac{2 - 2\cos^2\theta}{1 + \cos\theta}$
2. $\dfrac{\cos\theta - \sin\theta}{\cos^2\theta - \sin^2\theta}$

# IV- Double Angle Identities

**Property**

$\sin 2\theta = 2\sin\theta\cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$\cos 2\theta = 1 - 2\sin^2\theta$

$\cos 2\theta = 2\cos^2\theta - 1$

$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

**Example**

Given that $\sin\alpha = \frac{3}{5}$ and $\cos\alpha = -\frac{4}{5}$, find :

1. $\sin 2\alpha$
2. $\cos 2\alpha$
3. $\tan 2\alpha$

**Example**

If $\sin\alpha = \frac{5}{13}$ where $\frac{\pi}{2} < \alpha < \pi$, find the exact value of :

1. $\sin 2\alpha$
2. $\tan 2\alpha$

**Example**

If $\alpha$ is acute and $\cos 2\alpha = \frac{3}{4}$, find the exact value of :

1. $\cos\alpha$
2. $\sin\alpha$

**Example**

Use an appropriate double angle identity to simplify :

1. $3\sin\theta\cos\theta$
2. $4\cos^2 2B - 2$

# V- Compound Angle Identities

**Property**

or any two angles :

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

**Example**

Expand and simplify $\sin(270^\circ + \alpha)$.

**Example**

Simplify $\cos 3\theta \cos \theta - \sin 3\theta \sin \theta$.

**Example**

Without using your calculator, show that $\sin 75^\circ = \frac{\sqrt{6}+\sqrt{2}}{4}$.

**Example**

Suppose $\sin x - \sqrt{3} \cos x = k \cos(x + b)$ for $k > 0$ and $0 < b < 2\pi$. Find $k$ and $b$.